

Luis Fernando Bustos-Márquez

Curriculum Vitae

Education

- 2022–Present **Ph.D. Candidate in Electrical and Computer Engineering**, *University of Delaware* – Expected May 2027, DE, United States, GPA – 3.9/4.0
- 2020–2022 **Master of Science in Mathematics Education**, *Francisco de Paula Santander University*, Cucuta, Colombia, GPA – 4.8/5.0
- 2017–2018 **Specialist in University Pedagogical Practice**, *Francisco de Paula Santander University*, Cucuta, Colombia, GPA – 4.8/5.0
- 2010–2016 **Bachelor of Science in Electronics Engineering**, *Francisco de Paula Santander University*, Cucuta, Colombia, GPA – 4.2/5.0

Experience

- 05/2026–**Graduate Intern, Load Forecasting**, *Dominion Energy*, Glen Allen, VA, United States
08/2026 – Analyzed non-conforming and emerging load patterns to support utility load forecasting and planning
– Investigated data center load profiles and their implications for grid demand, forecasting uncertainty, and resource planning
– Applied data analytics and visualization workflows using utility-scale load datasets for data centers
– Collaborated with load forecasting engineers and planning stakeholders to translate analytical findings into operational insights
- 08/2023–**Teaching Assistant**, *University of Delaware, Dept. of Electrical and Computer Engineering*, Newark, DE, United States
Present – Led lab sessions, taught 4 lectures/semester, and developed course materials for 80+ senior and graduate students [ELEG457/657-Renewable Energy Microgrids]
– Led lab sessions, grading assignments, and guiding final projects for senior and graduate students [ELEG428/628-Solar Electric Technology and Applications]
- 08/2022–**Research Assistant**, *University of Delaware, Institute of Energy Conversion*, Newark, DE, USA
Present – Developed +250k renewable and resilient microgrid design to enable AI data centers in the United States
– Applied machine learning and data analytics to bifacial photovoltaic systems for energy production forecasting
– Modeled PV+Battery Systems for peak load shaving and resiliency utilizing Hardware-In-the-Loop
– Assisted the Faster-Than-Real-Time simulation and demonstration for resilient Distributed Energy Resources (DER) integration project in collaboration with the Department of Energy
– Mentor 3 ECE master's students in Machine Learning & database design for renewable energy projects
- 01/2019–**Adjunct Professor**, *Francisco de Paula Santander University*, Cucuta, Colombia
07/2022 – Teaching courses of Photovoltaic Solar Energy, Antenna Theory, Electronics I, Analysis of Electric Circuits in DC and AC, Microelectronics, Introduction to Industrial Automation, and Introduction to Electronics Circuits at the undergraduate program of Electronics Engineering
– Advised and supervised five undergraduate senior theses and capstone projects in renewable energy
– Co-authored the Educational Project for the Electronics Engineering Program (2020)
- 08/2017–**High School Teacher**, *London Bilingual Institute*, Cucuta, Colombia
07/2022 – Teaching Math and Physics. Chair of the Department of Mathematics. Leading high school research projects.
– The school reached the top-10 in Math tests at the state level in 2022

- 11/2017– **Junior Research Fellow**, *Francisco de Paula Santander University*, Cucuta, Colombia, Telecommunication Development Research Group
- Designed modular off-grid photovoltaic systems to power rural regions in Colombia
 - Statistical analysis of solar photovoltaic energy resources in rural regions of Norte de Santander, Colombia
 - Characterized and forecasted energy demand in non-interconnected regions in Norte de Santander, Colombia
 - Authored four technical reports and gave executive presentations to technical and non-technical audiences
- 08/2017– **Researcher**, *Francisco de Paula Santander University*, Cucuta, Colombia, Applied Microelectronics Development Research Group
- Research on the Analysis of the photovoltaic solar energy in rural areas of Norte de Santander and Characterization and projection of energy demand in non-interconnected areas of Norte de Santander
- 01/2017– **Support Professional**, *Francisco de Paula Santander University*, Cucuta, Colombia, Undergraduate program of Electronics Engineering
- Assisting the data analysis and planning activities for the high-quality accreditation in education
- 9/2016– **Support Engineer**, *Francisco de Paula Santander University*, Cucuta, Colombia, Applied Microelectronics Development Research Group
- Conducting research on Solar chargers for mobile phones

Research and Development Projects

- 2025–Present **Design of Renewable and Resilient Renewable Energy Microgrids to Enable Artificial Intelligence (AI) Data Centers**, *University of Delaware*, Newark, DE, USA, PhD Project
- 2025–Present **Seasonal Solar Congestion Management (SEASCOM)**, *Pecan Street, Delaware Electric Cooperative, and University of Delaware*, Newark, DE, USA, PhD Project
- 2024–2025 **Data Analysis of the Rear-to-Front Irradiance Ratios in Horizontal Single Axis Tracking Systems Across Varying Sky Conditions**, *University of Delaware*, Newark, DE, USA, PhD Project
- 2023–2024 **Bifacial Photovoltaics Gain Estimation Applying Machine Learning Methods**, *Institute of Energy Conversion, University of Delaware*, Newark, DE, USA, PhD Project
- 2022–2023 **Peak Load Shaving using Photovoltaics and Battery Energy Storage Systems**, *Department of Energy, Institute of Energy Conversion, University of Delaware*, Newark, DE, USA, PhD Project
- 2020–2022 **Analysis of the Saber Pro tests of the Electronics Engineering Program**, *Francisco de Paula Santander University*, Cucuta, Colombia, Master's thesis.
- 2017–2018 **Modular off-grid photovoltaic system to power non-interconnected areas of Norte de Santander**, *Francisco de Paula Santander University*, Cucuta, Colombia, Junior Research Fellow. Contract number: 062-2017
- 2017–2017 **Analysis of the photovoltaic solar energy in rural areas of Norte de Santander**, *Francisco de Paula Santander University*, Cucuta, Colombia, Researcher. Contract number: 017-0714
Sponsor: Francisco de Paula Santander University, Energy Mining Planning Unit and Institute of Planning and Promotion of Energy Solutions for Non-Interconnected Areas.
- 2017–2017 **Characterization and projection of energy demand in non-interconnected areas of Norte de Santander**, *Francisco de Paula Santander University*, Cucuta, Colombia, Researcher. Contract number: 017-0714
Sponsor: Francisco de Paula Santander University, Energy Mining Planning Unit and Institute of Planning and Promotion of Energy Solutions for Non-Interconnected Areas.
- 2016–2017 **Solar chargers for mobile phones**, *Francisco de Paula Santander University*, Cucuta, Colombia, Principal Researcher
- 2015–2016 **Modular off-grid photovoltaic system with charge controller based on microcontroller**, *Francisco de Paula Santander University*, Cucuta, Colombia, Undergraduate Thesis Project.

Honors and Awards

- 2026 **Best Poster Award.** Dominion Energy Virginia - Electric Transmission. Recognized for contributions during the summer internship program and winning the Best Poster Award
- 2026 **Outstanding Teaching Assistant Award.** Department of Electrical & Computer Engineering, University of Delaware, USA. Recognized for excellence in teaching, student support, and instructional contributions in electrical and computer engineering.
- 2026 **Student Grant Award.** Hispanics in Energy - Virginia (HIE-VA). Awarded for contributions advancing diversity, workforce readiness, and innovation in the energy sector.
- 2025 **Junior Researcher & National Peer Reviewer.** National Research Classification System. Ministry of Science, Technology, and Innovation (MinCiencias), Colombia. National classification for research excellence and peer-review contributions.
- 2025 **Best Poster Award.** Department of Electrical & Computer Engineering, University of Delaware, USA. "Powering Up AI Research Facilities with Resilient Renewable Energy Microgrids." Category: Nanoelectronics, Electromagnetics & Photonics.
- 2024 **Westerman Family Graduate Research Fund Fellowship.** University of Delaware, USA. For graduate research in climate change and renewable energy.
- 2024 **Best Poster Award.** IEEE 11th Conference on Technologies for Sustainability (SusTech 2024). Portland, OR, USA. "Energy Management Strategy and Solar Energy Utilization in Oversized Solar Arrays and Grid-Tied Photovoltaic Systems."
- 2023 **Best Poster Award.** Delaware Environmental Institute (DENIN)- Environmental Symposium. "Energy Management Strategy and Solar Energy Utilization in Oversized Solar Array and Grid-tied PV systems." For research on photovoltaic system optimization and energy management.

Selected Finalist Recognitions

- 2025 **Ph.D. Dissertation Challenge Finalist.** Oral Presentation. Peak Load Shaving using Residential Scale Photovoltaics and Battery Energy Storage Systems (Winter). IEEE PES Grid Edge Technologies Conference and Exposition, San Diego, CA, USA
- 2024 **Best Student Oral Presentation Award Finalist.** Front and Rear Irradiance in Bifacial PV systems. IEEE 52nd Photovoltaic Specialists Conference, Seattle, WA, USA

Academic and Professional Distinctions

- 2025 **Sigma Xi - Full member.** The Scientific Research Honor Society, USA
- 2024 **Tau Beta Pi Graduate Member.** Engineering Honor Society .University of Delaware, USA
- 2022 **Recognition of research work-Patent of Invention.** Francisco de Paula Santander University
- 2022 **Award for Entrepreneurial Merit.** Municipal Council of San José de Cúcuta, Colombia
- 2016 **National Top Score, Saber Pro Examination** Electronics Engineering Program, Colombia. Highest national score in the Electronics Engineering program in the areas of Scientific Thinking in Physical Sciences and Written Communication.
- 2016 **Francisco de Paula Santander University, Colombia Distinction,** Academic excellence scholarship awarded for high GPA during 80% of the undergraduate program.
- 2016 **Meritorious Undergraduate Thesis,** Universidad Francisco de Paula Santander, Colombia. "Modular Off-Grid PV System with Charge Controller Based on Microcontroller"

Selected Nominations

- 2024, 2026 **Graduate Student Excellence in Service & Teaching Award Nominee.** College of Engineering, University of Delaware, USA

Additional Training

- 2025 Preparing Future Faculty Workshop. *Auburn University*
- 2023-2025 Summer Institute of Teaching. *Center for Teaching and Assessment of Learning, UD*
- 2023-2025 Innovative Teaching Strategies *IT Academic Technology Services, University of Delaware*
- 2025 Responsible Conduct of Research. *Collaborative Institutional Training Initiative*
- 2025 Communications Training for UD Researchers - Cohort 2. *Research Office, University of Delaware*
- 2023 Design and optimization of Distributed Energy Resources (DER) for carbon reduction, resiliency, and return of the investment *Xendee Microgrid Simulation and Optimization Software*
- 2023 International Teaching Assistant (ITA) Program. Certified as Instructor Category I by *English Language Institute (ELI)*. University of Delaware

Service & Leadership

- 06/2025 **Senator, Graduate Student Government**, *PhD Senator*, Department of Electrical and Computer Engineering, University of Delaware, DE, United States
- 05/2026
 - Advocate for graduate students by representing the interests and concerns of the **ECE Department**
 - Voted on graduate student legislation and officer elections, contributing to policy and governance decisions
 - Research Advancement Committee member. Proposing legislation to advance research at UD
- 04/2026 **STEM Outreach Presenter**, *Energy Expo*, Skyline Middle School, Wilmington, Delaware
- 01/2023 **Graduates in Electrical and Computer Engineering, GECE**, *President 24'-25', VP 23'-24'*, Department of Electrical and Computer Engineering, University of Delaware, DE, United States
- 06/2025
 - Promoted graduate student community-building and collaboration within the ECE department.
 - Successfully leadership on 1 <3 UD fund raising campaign, 10+ research and social events, collaboration with 7+ UD departments and offices, strong communication with ECE Department.
- 01/2023 **Hispanic/Latino Graduate Student Association, HLGSA**, *Secretary*, University of Delaware, DE, United States
- 07/2023
 - Promoted Latino and Hispanic culture by strengthening community and belonging at the UD.
 - Monitored the board and checking administration supplies.
 - Planning of activities, such as Game Day, Cooking Events, and Journey to Ecuador
 - In 2023, HLGSA was in the top-3 of the biggest Graduate Student Organizations at UD.
- 2024-2025 **College and Professional Engagement**, COE-University of Delaware, DE, United States
- 2024-2025
 - Poster Presenter**, National Academy of Engineering (NAE) Spring Regional Meeting, University of Delaware, STAR Campus (2024)
 - Poster Presenter**, Reception for Dr. Susan Margulies (Assistant Director for Engineering, National Science Foundation), College of Engineering, University of Delaware (2024)
 - Panelist**, "Preparing for Graduate School" Session, Research Experience for Undergraduates (REU), College of Engineering, University of Delaware
- 05/2019 **Curriculum Committee**, *Alumni Representative*, Undergraduate Program of Electronics Engineering, Francisco de Paula Santander University, Cucuta, Colombia
- 06/2020
 - Co-author of the Educational Project of the Electronics Engineering Program (2020)
 - Training about college national level tests in Scientific Thinking in Physical Science (2019-2020)
- 02/2019 **IEEE Student Branch**, *Power and Energy Systems-Chapter Advisor*, Undergraduate Program of Electronics Engineering, Francisco de Paula Santander University, Cucuta, Colombia
- 12/2019
 - Organizing and planning the "2019 Solar Race", student solar scale cars competition.
 - Advising and promoting the utilization of solar energy at the college and state level

Patent

Bustos-Márquez L.F., & Sepúlveda-Mora S.B. (2021). Device and Method of charging and discharging a battery through Maximum Power Point Tracking. Francisco de Paula Santander University. Applied Microelectronics Development Research Group. Superintendence of Industry and Commerce. Registration number: NC2017/0008573, Classification IPC: H02J 7/00, H02J 7/02. From August 2017 to August 2032.

Peer-reviewed Journal Papers

- Bustos-Márquez, L.F.** & Hegedus S. (2026). Techno-Economic Analysis of Renewable Energy Microgrids to Power Up AI Data Centers in the USA. *Renewable Energy* ISSN: 0960-1481. In preparation for submission.
- Bustos-Márquez, L.F.** & Hegedus, S. (2026). An Algorithmic Framework for Plant-Level AC Power Estimation in a Bifacial Horizontal Single-Axis Tracking PV System Using Explainable and Ensemble Machine Learning. *Algorithms*, 19(6), 496. <https://doi.org/10.3390/a19060496>
- Bustos-Márquez, L.F.** & Hegedus S. (2026). Long-Term Assessment of Rear-to-Front Irradiance Variability and Its Implications for Bifacial PV Performance Monitoring. *Solar Energy*, 315, 114784. ISSN: 0038-092X. <https://doi.org/10.1016/j.solener.2026.114784>
- Bustos-Márquez L.F.**, Sepulveda-Mora S.B., & Gonzalez-Sepulveda R.M. (2026). Factors Associated with Performance on the Saber Pro Exams Using Data Analytics and Machine Learning. *Journal of Research, Development, and Innovation*. ISSN: 1900-8260. Accepted.
- Bustos-Márquez, L.F.**, Sepúlveda-Mora, S.B., Guevara-Ibarra, D., and Medina-Delgado B. (2023). MPPT charge controller with state of charge estimation method based on ampere-hour counting and open circuit voltage. *Ingeniería*. 28(2), e18722. <https://doi.org/10.14483/23448393.18722>.
- W. Contreras-Sepúlveda, M.G. Galbán-Pineda, **L. F. Bustos-Márquez**, S.B. Sepúlveda-Mora, and J.J. Ramírez-Mateus (2021). Angström-Prescott empirical model to estimate solar radiation in Norte de Santander, Colombia. *Revista de Investigación, Desarrollo e Innovación*, 11(2), 413–428. <https://doi.org/10.19053/20278306.v11.n2.2021.12765>.
- S.B. Sepúlveda-Mora, E.A. Luna-Paipa, M.A. Laguado-Serrano, **L. F. Bustos-Márquez** (2019). Performance comparison between PWM and MPPT charge controllers. *Scientia et technica*, vol. 24, no. 1, p.6, 2019, <https://doi.org/10.22517/23447214.20681>

Conferences

- Bustos-Márquez, L.F.** & Hegedus S. (2026). Evaluating Resilient Renewable Energy Microgrids for Data Centers in the USA. *2026 IEEE Transmission and Distribution Conference & Exposition*, Chicago, IL, USA. <https://doi.org/10.1109/TD48022.2026.11562863>
- Bustos-Márquez, L.F.** & Hegedus S. (2025). Analyzing Rear-to-Front Irradiance Ratios in Bifacial HSAT Systems Across Varying Sky Conditions. *2025 IEEE 53rd Photovoltaic Specialists Conference (PVSC)*, Montreal, Canada. <https://doi.org/10.1109/PVSC59419.2025.11133318>
- Bustos-Márquez L.F.** & Hegedus, S. (2025). Peak Load Shaving Utilizing Distributed Residential Photovoltaics and Battery Energy Storage Systems. In *IEEE PES Grid Edge Technologies*. San Diego, CA, January 2025.
- Bustos-Márquez, L.F.**, Kaewnukultorn, T., & Hegedus, S. (2024). Front and Rear Irradiance ratio in Bifacial Photovoltaic Systems. *2024 IEEE 52nd Photovoltaic Specialist Conference (PVSC)*, Seattle, WA, USA. <https://doi.org/10.1109/PVSC57443.2024.10749508>
- Bustos-Márquez L.F.**, Kaewnukultorn & Hegedus, S. (2024). Energy Management Strategy and Solar Utilization in Oversized Solar Array and Grid-tied Photovoltaic Systems. In *2024 IEEE SusTech Conference*. Portland, OR, USA, February 2024.
- Bustos-Márquez L.F.**, Kaewnukultorn & Hegedus, S. (2023). Energy Management Strategy and Solar Energy Utilization in oversized solar photovoltaic systems. In *2023 DENIN Environmental Symposium*. University of Delaware, DE, USA, April 2023.
- Bustos-Márquez L.F.**, Kaewnukultorn & Hegedus, S. (2023). Peak Load Shaving using Photovoltaics (PV) and Battery Energy Storage Systems (BESS). In *Research Day*. University of Delaware, DE, USA, May 2023.

- Bustos-Márquez L.F.**, Ramirez-Mateus J.J., Sepúlveda-Mora S.B., & Vega M.F. (2020). Smart Building in the "Aulas Generales" building of the Francisco de Paula Santander University. In *7th International Week of Science, Technology and Innovation*. Francisco de Paula Santander University (UFPS). Cucuta, Colombia.
- Bustos-Márquez L.F.**, Ramirez-Mateus J.J., Uribe D.Y., Laguado-Angel L.E., & Rangel-Rolon J.A. (2019). Bluetooth communication for the emulation of the control of a lighting system in Proteus 8.1. In *6th Int. Week of Science, Technology and Innovation*. Francisco de Paula Santander University. Cucuta, Colombia.
- Moreno-García F.E., Ramirez-Mateus J.J., Sepúlveda-Mora S.B., **Bustos-Márquez L.F.**, (2019). Off-grid photovoltaic system for a Peach Production Project in non-interconnected zone: case study. In *6th International Week of Science, Technology and Innovation*. UFPS. Cucuta, Colombia.

Invited talks

- Bustos-Márquez L.F.** (2026). AI and the Environment - Exploring Perspectives on the environmental impacts of AI expansion. In *The Delaware Environmental Institute (DENIN) Panel Series*. Panelist. University of Delaware
- Bustos-Márquez L.F.** (2026). Data Centers and Microgrids: Powering AI Sustainably. In *2026 Space Data Centers Summer Camp* Keynote Speaker. George Mason University.
- Bustos-Márquez L.F.** & Hegedus S. (2025). Hybrid Microgrids for Data Center On-Site Power. In *Senator Hansen's DE State Energy Stakeholder Panel*. Speaker.
- Bustos-Márquez L.F.**, Sepulveda-Mora S.B., & Gonzalez-Sepulveda R.M. (2024). Analysis of the National Standardized Test Saber Pro of the Undergraduate Program of Electronics Engineering of the Francisco de Paula Santander University Applying Machine Learning Techniques. In *XI International Week of the Science and Technology* Keynote Speaker. Education Seminar Series
- Bustos-Márquez L.F.** & Hegedus, S. (2023). Peak Load Shaving utilizing Photovoltaics and Battery Energy Storage Systems. In *Institute of Energy Conversion (IEC)*. Keynote Speaker. Seminar series.
- Bustos-Márquez L.F.** (2023). Batteries, the key to a more sustainable and greener electric grid. In *PITCH 90: Bi-annual environmental research pitching event* Presenter. Delaware Environmental Institute (DENIN), University of Delaware. Newark, DE.
- Bustos-Márquez L.F.** (2023). Research results visualizations utilizing Python. In *UFPS-Research Training Series* Keynote Speaker. Francisco de Paula Santander University. Cúcuta, Colombia.

Software

- Guevara-Ibarra D., Medina-Delgado B., **Bustos-Márquez L.F.**, Beltrán-Ortega P.A., Mendoza-García J.L., & Rolon-Quintero D.C. (2019). MAPRNI. Francisco de Paula Santander University. Telecommunication and Electronics Development Research Group. Registration-National Directorate of Copyright (DNDA): 13-71-336.
- Bustos-Márquez L.F.**, Sepúlveda-Mora S.B., & Ramirez-Mateus J.J. (2016). MPPT charge control algorithm with Incremental Conductance method. Francisco de Paula Santander University. Applied Microelectronics Development Research Group. Registration number - National Directorate of Copyright (DNDA): 13-60-165.
- Bustos-Márquez L.F.**, Sepúlveda-Mora S.B., & Ramirez-Mateus J.J. (2016). MPPT charge control algorithm with Perturb and Observe method. Francisco de Paula Santander University. Applied Microelectronics Research Group. Registration - National Directorate of Copyright (DNDA): 13-60-165.

Computer skills

Advanced HOMER Grid, SAM, Matlab, Python, Microsoft Office, Orcad
Intermediate L^AT_EX, Xendee, C++, R

Languages

Spanish Native
English Advanced, Certified by IELTS, TOEFL, and ITA UD

Interests

- Renewable Energy Microgrids
- Photovoltaics
- Machine Learning
- Energy Storage Systems

Professional Memberships

2016–Present **Colombian Association of Engineers, ACIEM**, *Professional Member*.

2016–Present **Institute of Electrical and Electronics Engineers, IEEE**, *Professional Member*, Power & Energy Society member, IEEE Young Professionals, Smart Grid Community.